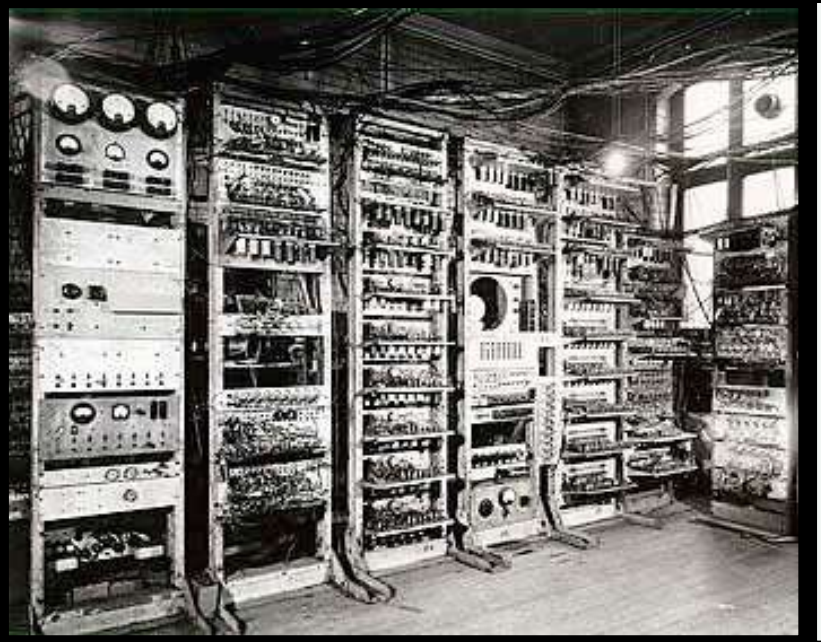


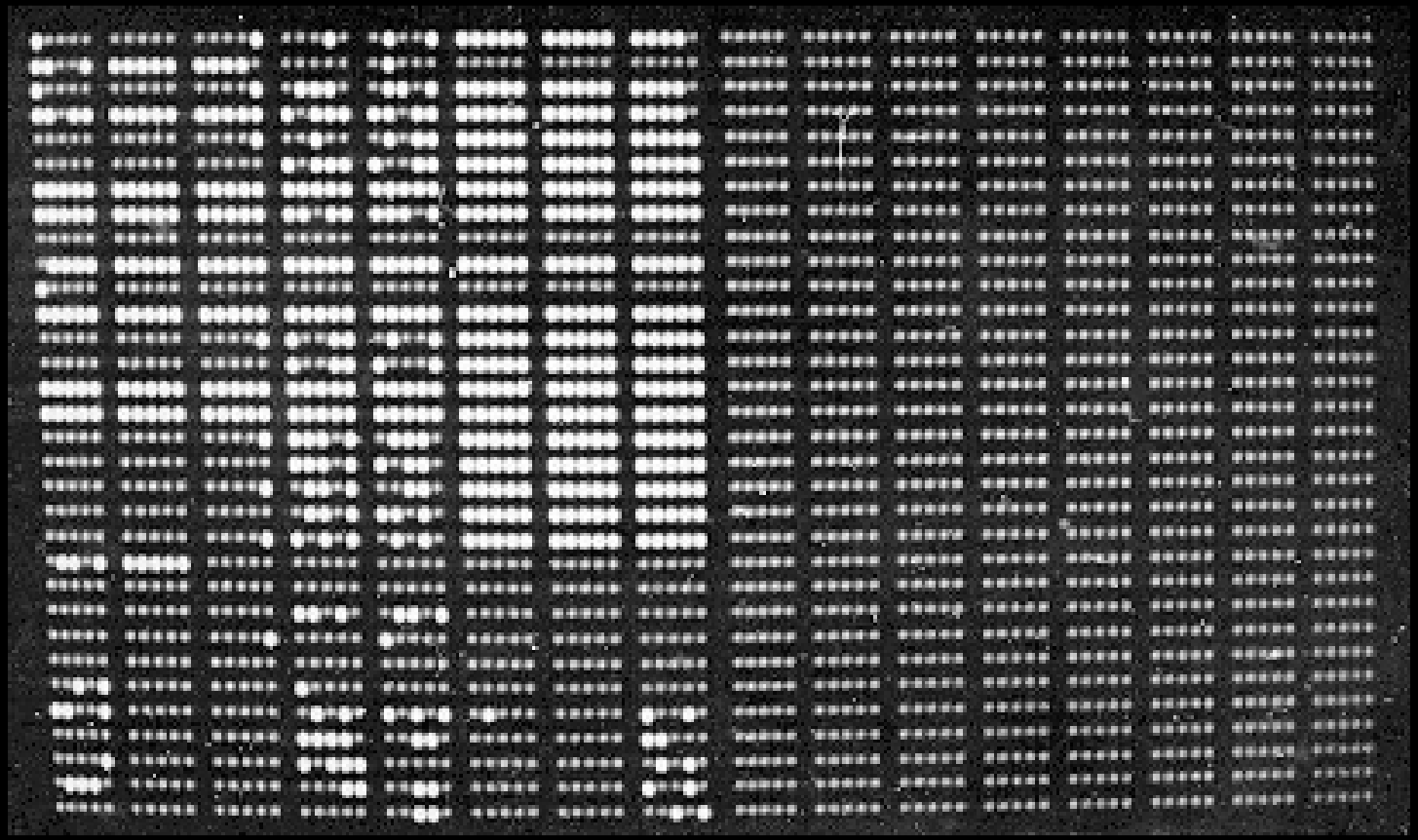

# IMPORTÂNCIA DA COMPUTAÇÃO GRÁFICA

JOAQUIM JORGE, RUI PRADA

BASEADO EM SLIDES ORIGINAIS DE CARLOS MARTINHO

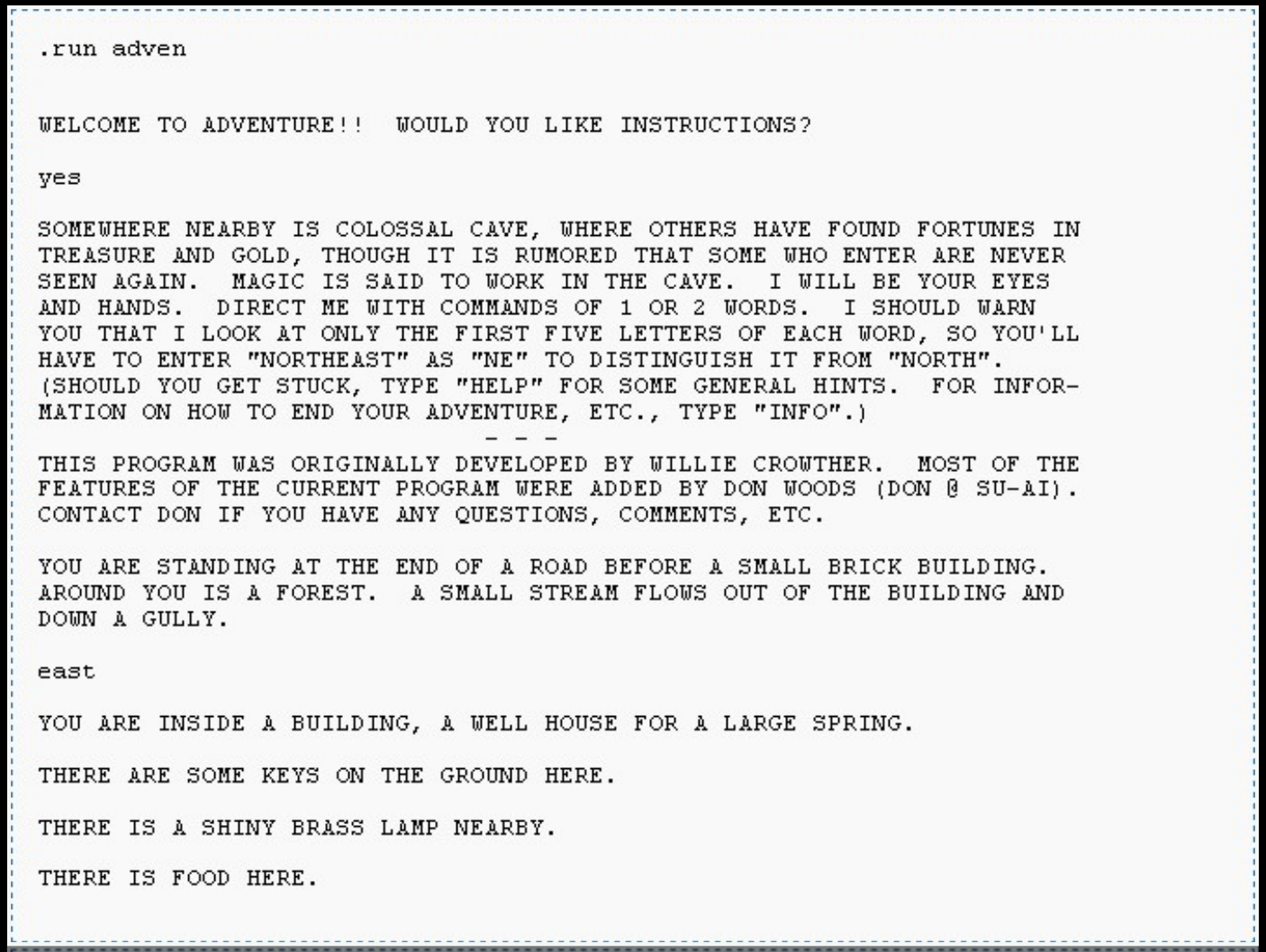






*Necessidade de **sintetizar** imagens num dispositivo de **visualização** a partir de um modelo de **cena (informação)***

# *Computação Gráfica*




Colossal Cave Adventure (Crowther, 1975)




Colossal Cave Adventure (Level 9, 1986)

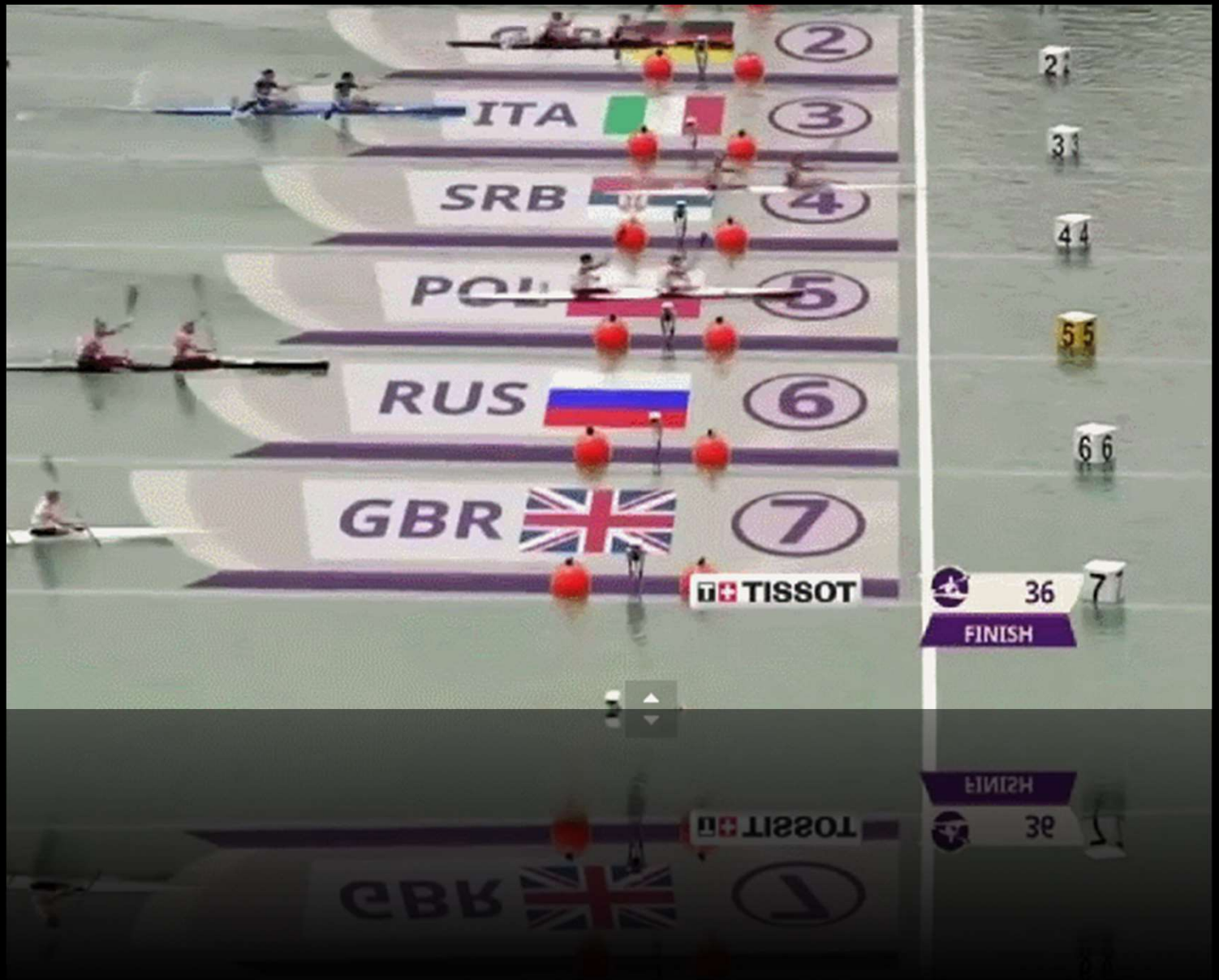



Secret of Monkey Island (Lucasfilms Games, 1992)




(Naughty Dog, SCE, 2013)


facebook
YouTube
flickr
LIVE
A12
E25
98 KPH
100 MAX
80 METERS
12:31
2:30 HRS
TOMTOM


2
ITA
3
SRB
4
5
RUS
6
GBR
7
TISSOT
36
FINISH



Ni No Kuni (Level-5, Bandai Namco Entertainment, 2012 release)



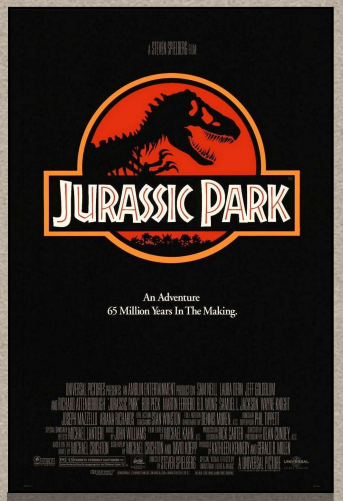
# COMPUTAÇÃO GRÁFICA EM FILMES



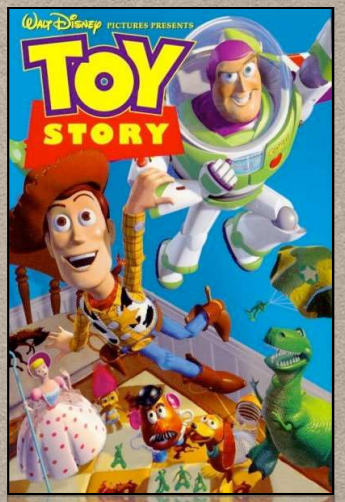

(1982)



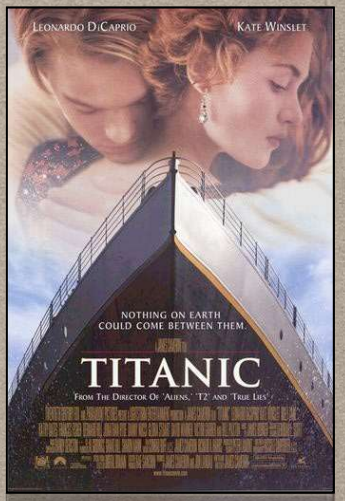

(1991)




(1993)




(1995)




(1997)



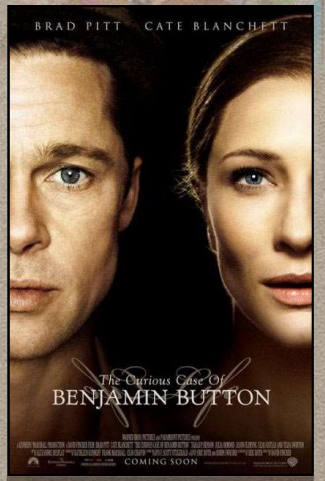

(1999,2002,2005)



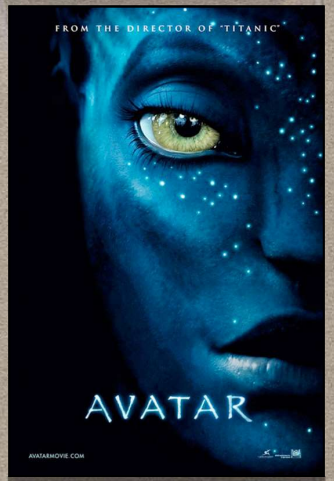

(1999)



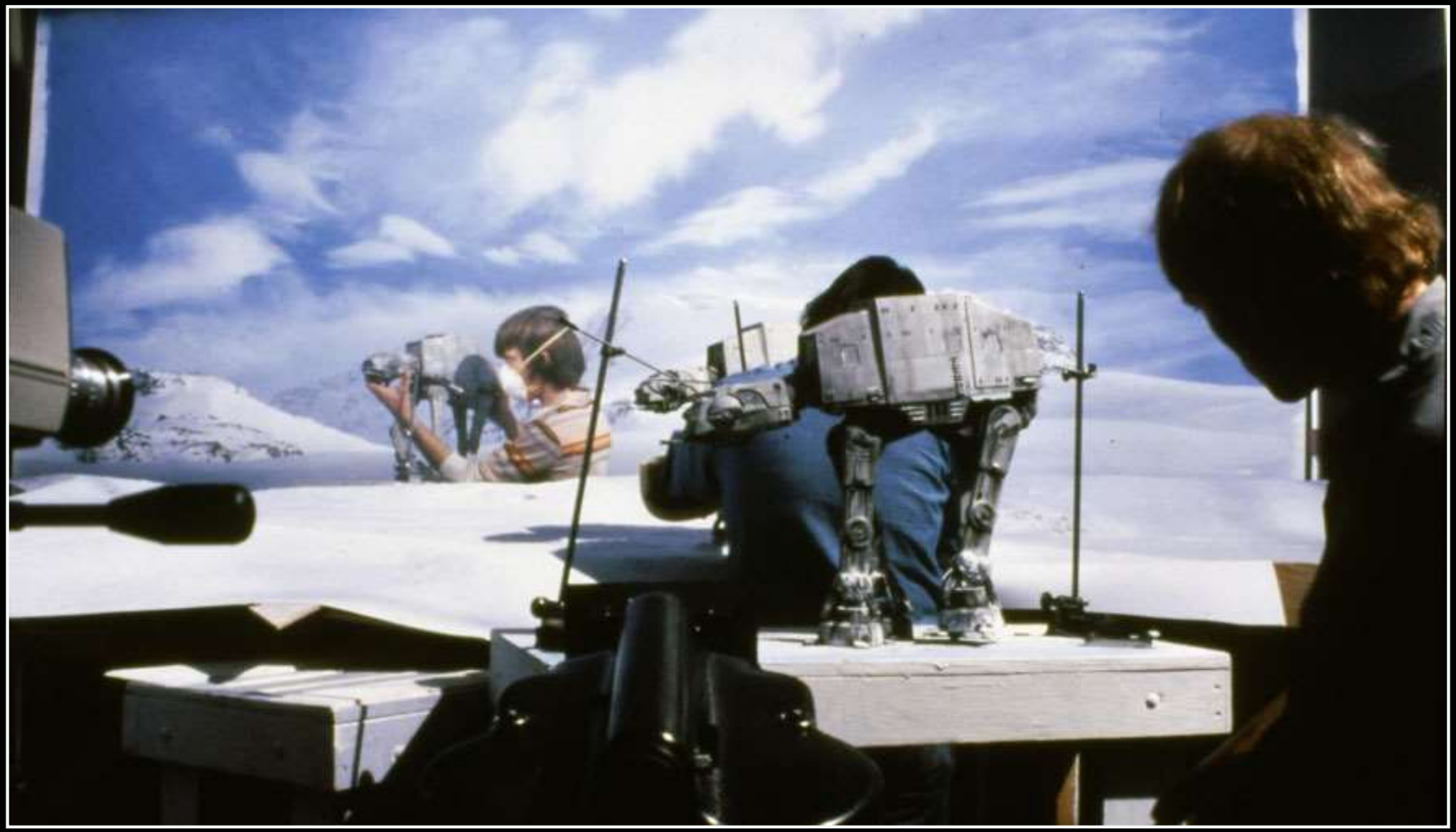

(2001, 2002, 2003)




(2008)




(2009)



Star Wars Episode V (Industrial Light & Magic)


A world inside
the computer
where man
has never been.
Never before now.
TRON
Enter its world this summer.
TRON A LISBERGER-KUSHNER PRODUCTION
STARRING JEFF BRIDGES BRUCE BOXLEITNER DAVID WARNER CINDY MORGAN AND BARNARD HUGHES
EXECUTIVE PRODUCER RON MILLER MUSIC BY WENDY CARLOS STORY BY STEVEN LISBERGER AND BONNIE MACBIRD
SCREENPLAY BY STEVEN LISBERGER PRODUCED BY DONALD KUSHNER DIRECTED BY STEVEN LISBERGER
SONGS BY JOURNEY FROM WALT DISNEY PRODUCTIONS
TECHNICOLOR®
FILMED IN SUPER PANAVISION® 70
DOLBY STEREO™ Read the Ballantine Book Original motion picture sound track available CBS Records and Tapes
© 1982 Walt Disney Productions

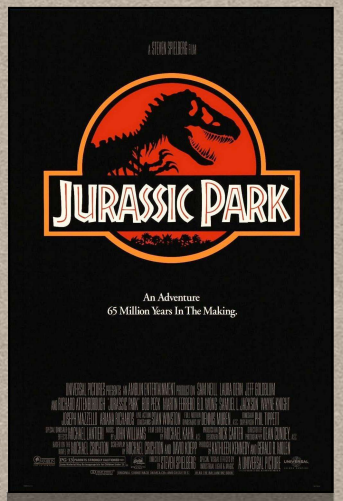
# COMPUTAÇÃO GRÁFICA E FILMES



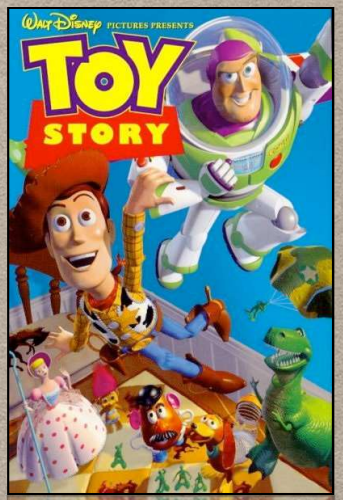

(1982)



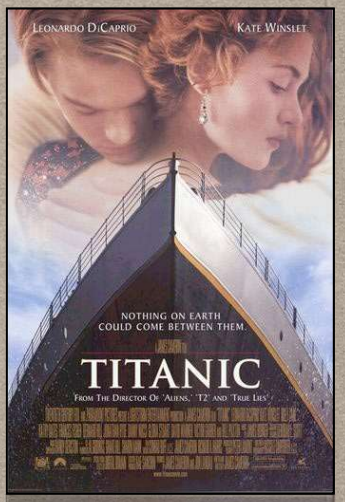

(1991)




(1993)




(1995)




(1997)




(1999,2002,2005)



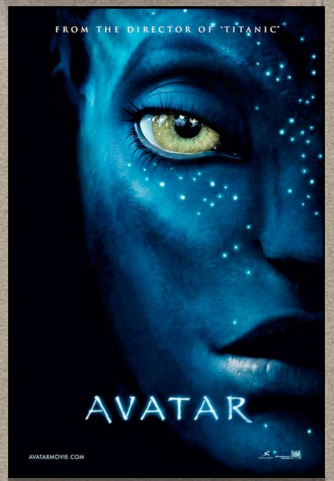

(1999)




(2001, 2002, 2003)




(2008)




(2009)



GDC 2016 - Live Performance in Unreal Engine 4
Hell Blade (Ninja Theory)

# COMPUTAÇÃO GRÁFICA E JOGOS

70 M




april 2016




50 M

10 M





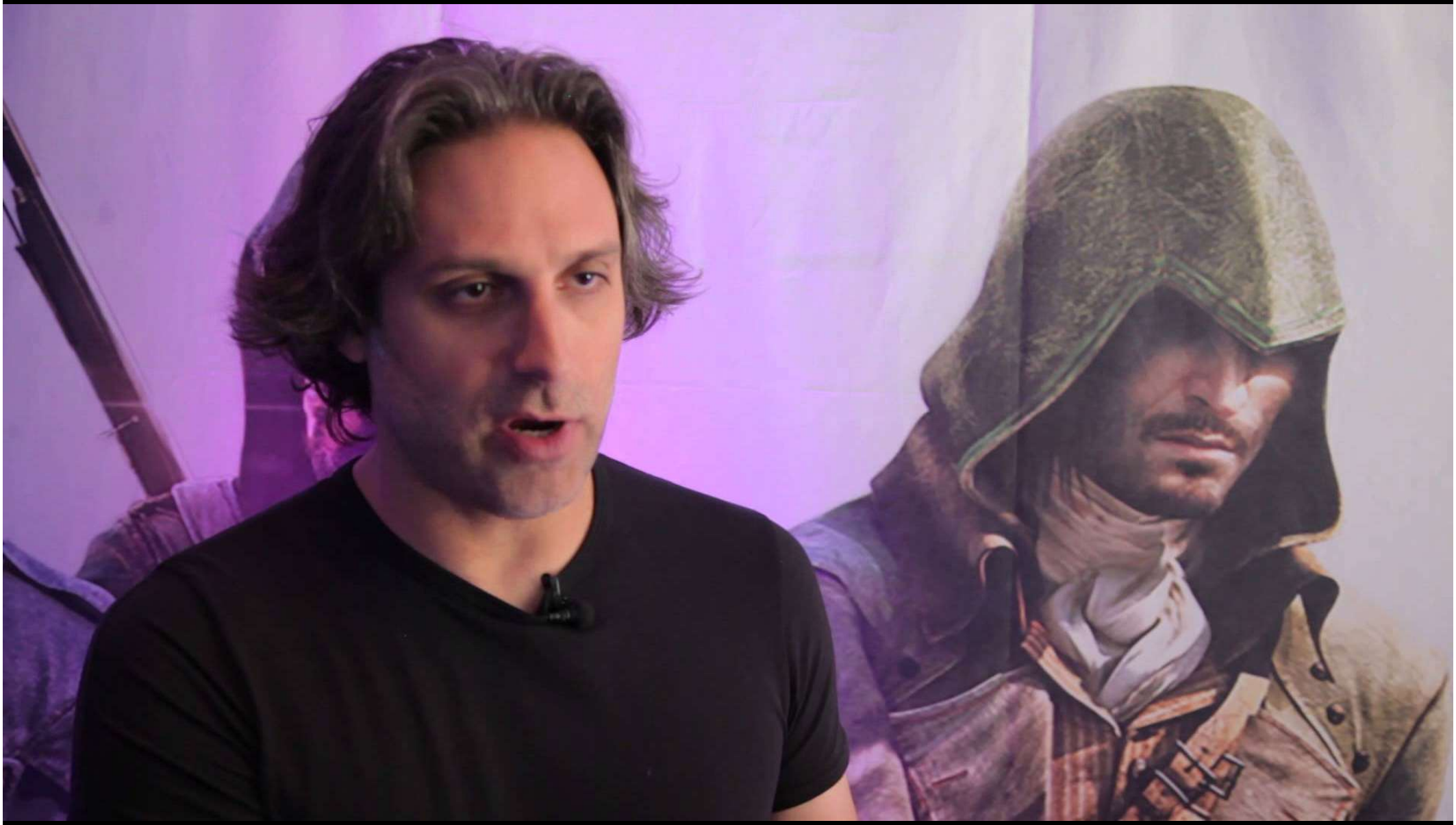


5.5 M

3 M




SONY
PS4



30 FPS “cinematic feel”
Alex Amancio, Creative Director, Ubisoft

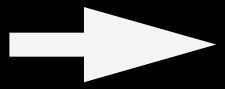
# SCREEN TEARING




Digital Foundry (Youtube oct. 2016)
Assetto Corsa (Kunos Simulazioni 2014)

# SINGLE BUFFERING





BUFFER

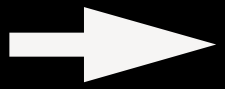




CENA
Geometria
Materiais
Luzes

# DOUBLE BUFFERING





BACK BUFFER



FRONT BUFFER



CENA
Geometria
Materiais
Luzes

VSYNC



# TRIPLE BUFFERING



BACK BUFFER

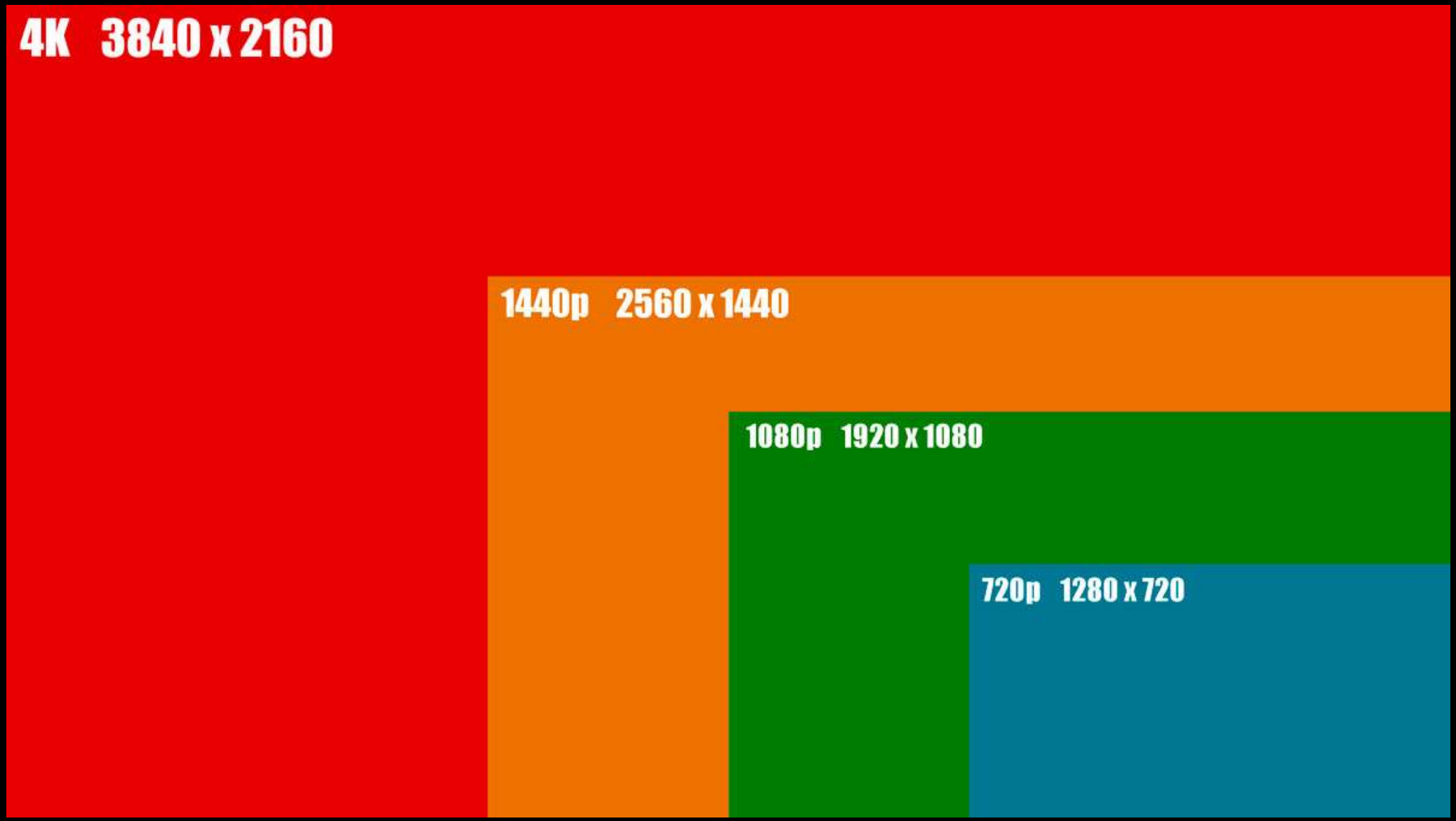
BACK BUFFER

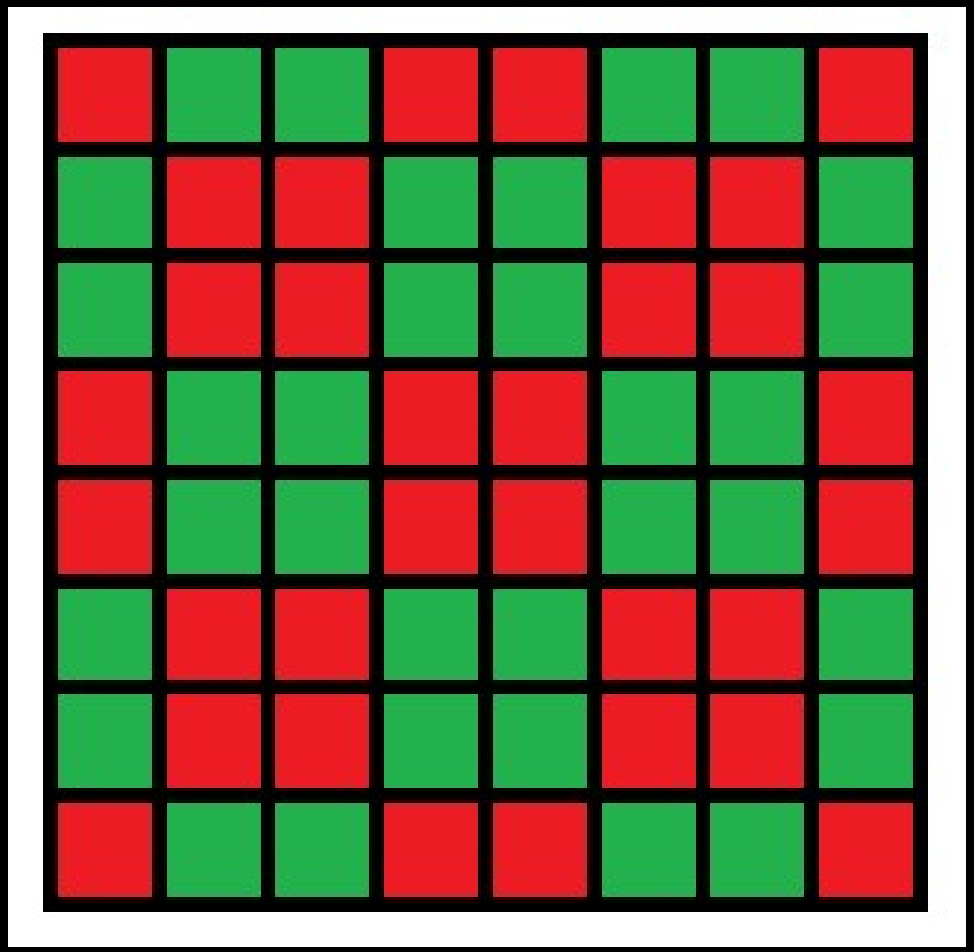
FRONT BUFFER

CENA
Geometria
Materiais
Luzes




NVIDIA


4K 3840 x 2160
1440p 2560 x 1440
1080p 1920 x 1080
720p 1280 x 720



# CHECKERBOARD RENDERING



Digital Foundry (Youtube oct. 2016)
Shadow Warrior 2 (Flying Wild Hog, Devolver Digital, 2016)



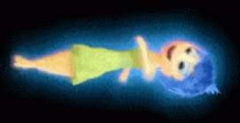

# Geometry

# Animation

# Rendering

# Imaging

# Topology






MINICLIP
PLAY GAMES

bica
STUDIOS




Hell Keeper (https://goo.gl/cTPJHB)



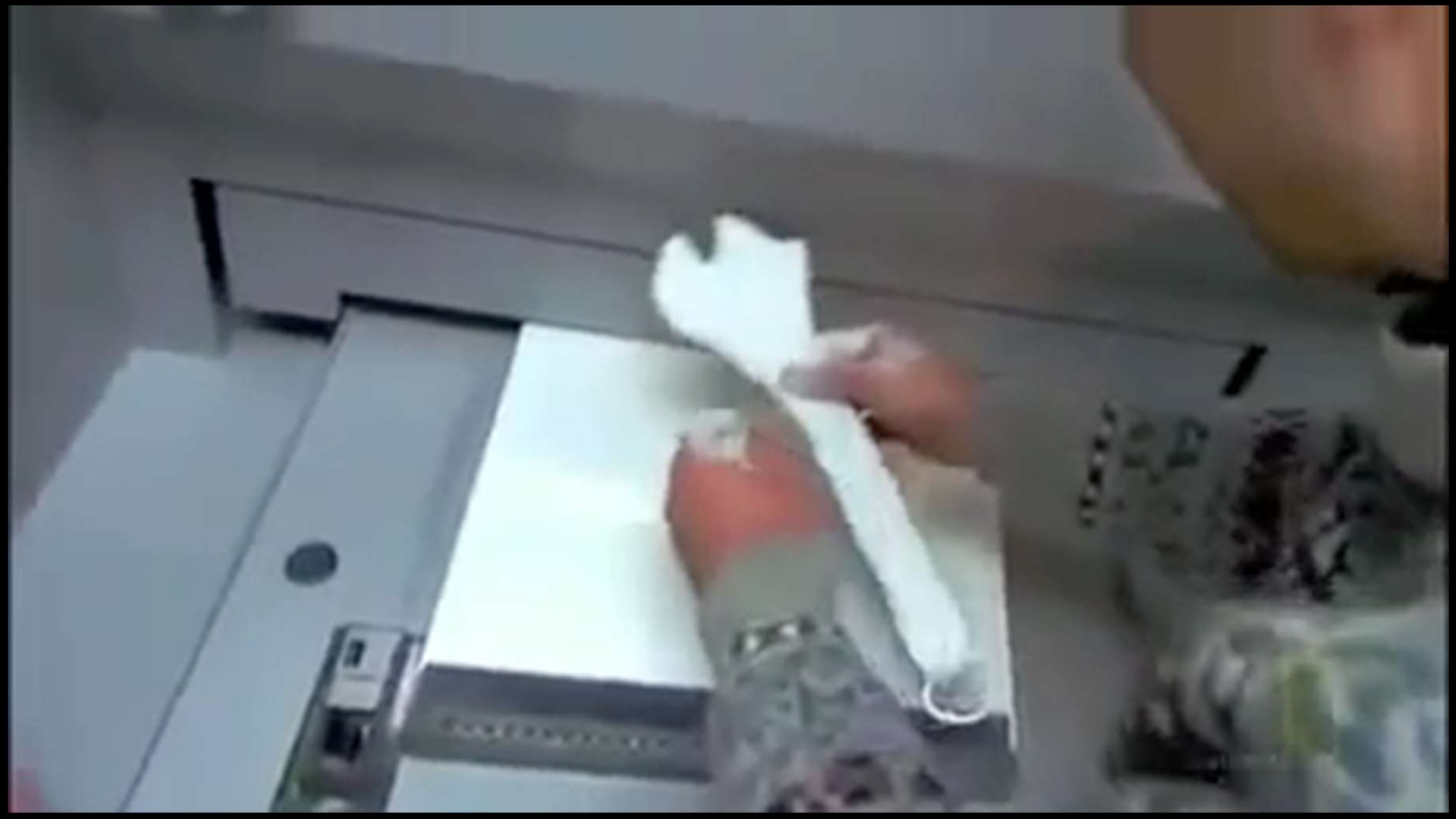

PlayStation.VR


oculus




UNREAL PARIS
VIRTUAL TOUR
Unreal Engine 4 Demo 2014





# 3D SCANNING AND PRINTING